

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech

Branch : AIDS/AI(AI & DS)/CSE(AI&DS)

Semester :III

Subject Code & Name: BTAIC303 Data Structure and Algorithm using Python

Max Marks: 60

Date:10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	How is a code block indicated in Python?	(L1/CO1)	1
	a. Brackets b. Indentation c. Key d. None		
2	What will be the output of the following code ? a=3 b=1 print(a,b) a,b=b,a print(a,b)	(L3/CO1)	1
	a. 3 1 1 3 b. 3 1 3 1 c. 1 3 1 3 d. 1 3 3 1		
3	Which of the following blocks will always be executed whether an exception is encountered or not in a program?	(L2/CO2)	1
	a. try b. except c. finally d. catch		
4	What is fruitful function in Python	(L4/CO2)	1
	a. function which will return value b. function which will not return value c. function which will send value. d.none		
5	What will be the Postfix expression for (a+b+c)*d	(L5/CO3)	1
	a. ab+c+d* b. ab+cd+* c. abc++cd* d. None		
6	Data Structure is Classified into :	(L1/CO3)	1
	a. Linear b. Non-Linear c. Non-Primitive d. All of these		

7	What is a hash table?				(L1/CO4)	1
	a. A structure that maps values to keys	b. A structure that maps keys to values	c. A structure used for storage	d. A structure used to implement stack and queue		
8	Which of the following is not a collision resolution technique?				(L2/CO4)	1
	a. Separate chaining	b. Linear probing	c. Quadratic probing	d. Hashing		
9	For a binary search algorithm to work, it is necessary that the array (list) must be				(L2/CO5)	1
	a. Sorted	b. Unsorted	c. In Heap	d. Popped out of stack		
10	Binary Search can be categorized into which of the following?				(L2/CO5)	1
	a. Brute Force technique	b. Divide-and-conquer	c. Greedy method	d. Dynamic programming		
11	Which sorting is also called divide and conquer sorting				(L2/CO5)	1
	a. Quick sort	b. Bubble sort	c. Merge sort	d. Both a & c		
12	What will be the output of the following code. dict1 = {'first': 'sunday', 'second': 'monday'} dict2={1:3, 2:4} dict1.update(dict2) print(dict1)				(L3/CO1)	1
	a. { 'first': 'sunday', 'second': 'monday', 1:3, 2:4}	b. {'first': 'sunday', 'second': 'monday'}	c. {1: 3, 2: 4}	d. None of these		
Q. 2	Solve the following.					12
A)	Explain features of Python in detail.				(L1/CO1)	6
B)	Explain concept of stack with its operations.				(L2/CO3)	6
Q.3	Solve the following.					12
A)	Define tree data structure. Explain different types of tree with examples				(L1/CO4)	6
B)	Write an algorithm for bubble sort? Sort following elements with bubble sir 45,67,23,13,20,10				(L2/CO5)	6
Q. 4	Solve Any Two of the following.					12
A)	Explain concept of Inheritance and its types in python with syntax.				(L2/CO2)	6

B)	Construct a Binary search tree for following data and perform in order, preorder and postorder 20,3,13,16,30,56,10,24,56	(L3/CO4)	6
C)	Write a python program to search element using Binary search method	(L2/CO5)	6
Q.5	Solve Any Two of the following.		12
A)	Write an algorithm for quick sort and insertion sort method.	(L2/CO5)	6
B)	Write a python program to construct BST and perform Preorder tree traversal.	(L3/CO4)	6
C)	Explain following a. Abstraction b. encapsulation c. class	(L1/CO2)	6
Q. 6	Solve Any Two of the following.		12
A)	Convert following infix expression to postfix : a. $(x+y)^2 + (a+b)^2$ b. $(p + q - r) * (s + t)$	(L3/CO3)	6
B)	Explain Linked list and its types with representation.	(L2/CO3)	6
C)	Write a python program to perform arithmetic operations using function.	(L2/CO2)	6
	*** End ***		